📄 Product Requirement Document (PRD)

📌 1. Document Control & Overview

• Product Name: Medify
• Target Launch: Q3 2027
• Platform: Android (Min SDK 26 / Android 8.0)
• Author: [Your Name] (Product Manager)
• Status: Draft / Ready for Engineering Review

1.1 Executive Summary

Medify is a holistic, AI-powered wellness Android application designed to make mindfulness and physical yoga practice deeply personalized and accessible. By leveraging Gemini AI for automated daily wellness summaries and integrating interactive 3D assets for yoga posture alignment, Medify bridges the gap between static content apps and real-world studio coaching.

1.2 Core Value Proposition

To provide an immersive, calming space for physical and mental wellness, anchored by actionable AI insights, interactive 3D learning, and a nature-inspired Green Aesthetic UI that scientifically induces stress reduction.

🎯 2. Personas & User Problems

2.1 Target Personas

1. "The Busy Professional" (Anxious Alex, 28): Struggles with high stress and desk fatigue. Needs quick, effective 10-minute meditation sessions with calming audio to reset during work breaks.
2. "The Remote Yogi" (Fitness Fiona, 34): Wants to practice yoga at home but worries about bad posture. Finds standard 2D video tutorials inadequate for understanding structural depth and alignment.

2.2 User Problem Statements

• “I watch yoga videos, but I can't tell if my back is straight or if my leg is angled correctly because the camera angle is fixed.”
• “I track my meditation and mood across different logs, but it’s just raw data—I don’t know what it means or how to improve my habits.”

🛠️ 3. Epics & Functional Requirements

Epic 1: The AI-Powered "Gemini Summary" Hub

Description: Consolidates a user's daily activity (meditation duration, yoga poses completed, and self-reported mood metrics) into a bite-sized, empathetic textual summary.
• FR-1.1: The system must collect data points (Meditation minutes, specific yoga poses completed, ambient sounds preferred, and user mood rating 1-5).
• FR-1.2: The app must securely send a structured JSON payload to the Gemini API via Google AI Studio integration.
• FR-1.3: The UI must display the AI response in a dedicated card component titled "Your Daily Mindful Digest" with actionable optimization tips (e.g., "You practiced Tree Pose today but noted feeling fatigued; try a grounding 5-min Forest soundscape before bed").

Epic 2: Interactive 3D Yoga Tutorial Engine

Description: Replaces traditional 2D videos with real-time manipulable 3D avatars demonstrating correct alignment.
• FR-2.1: The app must render 3D human models demonstrating at least 10 core yoga poses (e.g., Downward Dog, Warrior II, Tree Pose).
• FR-2.2: The UI must support standard touch gestures: 360-degree rotation (horizontal swipe) and Zoom in/out (pinch) so users can examine joint placements from all angles.
• FR-2.3: Bones/Joint Highlights: Key structural focal points (e.g., knees, spine) should glow subtly green when alignment warnings or tips are selected.

Epic 3: Immersive Meditation Timer & Calming Audio Mixer

Description: A distraction-free timer accompanied by multi-channel audio overlays.
• FR-3.1: Timer Interface: A clean, minimal circular countdown timer allowing custom duration inputs (1 to 120 minutes).
• FR-3.2: Audio Overlay System: Users must be able to toggle background ambient sounds (options: Rainforest Rain, Ocean Waves, Tibetan Singing Bowls, White Noise).
• FR-3.3: Background audio must continue playing smoothly if the device screen locks or goes into sleep mode (Android Foreground Service).

🎨 4. Design & UI/UX Specifications

4.1 The Visual Language: Eco-Wellness Green

To evoke tranquility and alignment with nature, Medify utilizes a 60-30-10 color rule heavily weighted towards organic greens:

• Dominant Color (60%): Mint/Sage Green (#E8F5E9 or #F0F7F4) for backgrounds—reducing visual strain.
• Secondary Color (30%): Deep Forest Green (#1B5E20 or #2C5E43) for text, primary buttons, and navigational elements to establish authority and grounding.
• Accent Color (10%): Light Emerald (#81C784) for active status indicators, timer highlights, and successful AI generation alerts.

4.2 Key Screen Wireframe Layouts

• Dashboard Screen: Top prominent card displaying the dynamic Gemini AI Summary. Below it, two horizontal scrolling feeds split into "Quick Meditation Presets" and "Explore 3D Poses".
• 3D Yoga Studio Screen: The 3D canvas occupies the top 60% of the viewport. Bottom sheet contains step-by-step text instructions, breathing tempos, and anatomical focus areas.

📊 5. Non-Functional Requirements & Metrics

5.1 Non-Functional Requirements (NFRs)

• Performance: 3D rendering engine must maintain a steady 60 FPS on mid-range Android devices. Gemini API latency must not exceed 2.5 seconds for response generation.
• Security: User mood metrics and personal logs must be stored locally using Android Room Encrypted Database or securely hashed if synced to cloud infrastructure.

5.2 Product Success Metrics (KPIs)

• North Star Metric: Weekly Active Sessions per User (Goal: Target ≥ 4 sessions/week indicating habit formation).
• Feature Adoption: Percentage of users who click "Expand 3D view" during a yoga flow.
• AI Retention Factor: Rate of users interacting with the Gemini summary daily versus skipping/dismissing it